

Technical Remediation Runbook

Detailed remediation procedure with command blocks and validation checks.

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1. Remediation Method

Each procedure below is written as an implementation-ready work instruction. Commands must be reviewed against the target environment before execution.

2. Technical Procedures

2.1 Apache Tomcat AJP Request Injection and Potential RCE

Affected Asset	10.20.1.234 / web-prod-01 / TCP 8009
CVE	CVE-2020-1938
Reference Links	https://tomcat.apache.org/security-9.html https://nvd.nist.gov/vuln/detail/CVE-2020-1938

Steps to Remediate

- Confirm whether the affected host is running a vulnerable Tomcat release and whether AJP is listening on TCP 8009.
- If AJP is not required, disable the AJP connector in server.xml and restart Tomcat.
- If AJP is required, upgrade Tomcat to a fixed release and restrict AJP to a trusted interface or loopback address.
- Configure an AJP shared secret when AJP must remain enabled, and align the reverse proxy connector with the same secret.
- Confirm TCP 8009 is not reachable from untrusted networks after the change.

Commands

Identify Tomcat version and AJP listener
<pre># Run on the Tomcat host export CATALINA_BASE=\${CATALINA_BASE:-/opt/tomcat} export CATALINA_HOME=\${CATALINA_HOME:-/opt/tomcat} \$CATALINA_HOME/bin/version.sh sudo ss -lntp grep ':8009' true sudo grep -R "protocol=\"AJP\" \"\$CATALINA_BASE/conf\" \"\$CATALINA_HOME/conf\" 2>/dev/null</pre>
Disable AJP when not required
<pre># Back up the config, comment out the AJP Connector in server.xml, then restart sudo cp "\$CATALINA_BASE/conf/server.xml" "\$CATALINA_BASE/conf/server.xml.bak.\$(date +%F)" sudo vi "\$CATALINA_BASE/conf/server.xml" sudo systemctl restart tomcat sudo systemctl restart tomcat9</pre>

Validation

- Run a new authenticated vulnerability scan and confirm the CVE-2020-1938 finding is closed.
- Confirm TCP 8009 is reachable only from the approved reverse proxy or local host.
- Attach the changed server.xml and Tomcat version output to the change ticket.

2.2 SSL Certificate Cannot Be Trusted

Affected Asset	10.20.1.235 / vpn-gw-02 / TCP 443
CVE	Not provided
Reference Links	https://www.tenable.com/plugins/nessus/51192 https://docs.tenable.com/whitepapers/useful-plugins/Content/UsefulPlugins/Resolving51192.htm https://docs.openssl.org/3.0/man1/openssl-s_client/ https://docs.openssl.org/3.0/man1/openssl-verify/

Steps to Remediate

- Identify the service, hostname, port, certificate subject, issuer, expiry, and chain presented by the affected asset.

- If the certificate is self-signed or issued by an internal CA, either replace it with an approved publicly trusted certificate or add the internal CA to the scanner trust configuration where appropriate.
- If the certificate chain is incomplete, install the required intermediate certificates on the service.
- If the certificate is expired or has the wrong subject/SAN, request and deploy a corrected certificate from the approved CA.
- Rerun the scan after the certificate chain validates cleanly.

Commands

Collect certificate and chain from the service

```
# Replace DNS and port with the affected service from the CSV
openssl s_client -connect vpn-gw-02:443 -servername vpn-gw-02 -showcerts </dev/null > vpn-gw-02_chain.txt
openssl x509 -in vpn-gw-02_chain.txt -noout -subject -issuer -dates -ext subjectAltName
```

Verify certificate chain with an approved CA bundle

```
# Use the approved enterprise/public CA bundle path for the target environment
openssl verify -CAfile approved-ca-bundle.pem vpn-gw-02_leaf.pem
```

Validation

- Confirm OpenSSL chain verification returns OK using the approved CA bundle.
- Confirm the service presents the complete chain after restart or reload.
- Rerun Tenable and confirm plugin 51192 is no longer reported for the asset/port.

2.3 Microsoft SQL Server Unsupported Version Detection

Affected Asset	10.20.2.118 / sql-prod-03 / TCP 1433
CVE	Not provided
Reference Links	https://learn.microsoft.com/en-us/sql/sql-server/end-of-support/sql-server-end-of-support-overview https://learn.microsoft.com/en-us/troubleshoot/sql/releases/find-my-sql-version https://learn.microsoft.com/en-us/troubleshoot/sql/releases/download-and-install-latest-updates https://learn.microsoft.com/en-us/sql/database-engine/install-windows/upgrade-sql-server

Steps to Remediate

- Confirm the SQL Server version, edition, product level, and build number.
- Compare the detected build with Microsoft lifecycle and latest-update guidance.
- Back up all affected databases and verify the backup before upgrade or migration.
- Choose the remediation path: in-place upgrade, side-by-side migration, Azure SQL migration, or Extended Security Updates when eligible.
- Patch to a supported SQL Server build and rerun validation queries after upgrade.

Commands

Identify SQL Server version and edition

```
-- Run in SSMS or sqlcmd against the affected instance
SELECT @@VERSION;
SELECT
    SERVERPROPERTY('ProductVersion') AS ProductVersion,
    SERVERPROPERTY('ProductLevel') AS ProductLevel,
    SERVERPROPERTY('Edition') AS Edition;
```

Back up and verify before remediation

```
-- Replace DBName and path for the target environment
BACKUP DATABASE [DBName]
TO DISK = 'D:\Backup\DBName_pre_upgrade.bak'
WITH CHECKSUM, INIT;

RESTORE VERIFYONLY
FROM DISK = 'D:\Backup\DBName_pre_upgrade.bak'
WITH CHECKSUM;
```

Validation

- Confirm the version/build maps to a supported Microsoft lifecycle state.
- Confirm application smoke tests and database jobs complete successfully.
- Rerun the vulnerability scan and confirm the unsupported SQL Server finding is closed.